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Preparation of $LiNbO_3$ thin film on insulator for high-performance lithium niobate devices

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ABSTRACT

The combination of ion-slicing and direct wafer bonding technology is a very promising method for transferring thin film material to cheap substrates. In this study, the evolution of internal defects and the thin film exfoliation process of He ion-implanted into $LiNbO_3$ crystal were studied based on ion-slicing technology. Then, based on the Ar plasma-activated bonding process, the $LiNbO_3$ thin film was transferred to the amorphous SiO_2 (a-SiO_2) layer. In addition, the bonding model of $LiNbO_3$/a-SiO_2/Si was established through finite element analysis, and the distribution curves of deformation and interfacial thermal stress with annealing temperature were obtained. After wet etching and proton exchange processes, the prepared LNOI substrate was successfully etched into a Y-shaped branching waveguide on the surface. The LNOI substrate prepared by this work can meet the practical application requirements of thin film $LiNbO_3$ devices.

1. Introduction

Lithium niobate ($LiNbO_3$, LN) has been widely used in integrated optics due to its excellent electro-optical, acoustic-optic, nonlinear optical properties, and wide transparency range [1]. The development of ultra-compact photonic integrated devices and circuits based on $LiNbO_3$ thin films is attracting more and more attention [2]. At present, there are many methods for preparing $LiNbO_3$ thin films, such as evaporation, sputtering, and epitaxy [3]. However, due to mismatches in crystal constants and thermal expansion coefficients between the $LiNbO_3$ film and the handling substrate, the quality of the $LiNbO_3$ film is significantly reduced [4]. In recent years, the fabrication of monocrystalline $LiNbO_3$ thin films has been realized using ion-slicing and wafer bonding techniques [5–8]. The $LiNbO_3$ on insulator (LNOI) substrates have provided a material platform for a variety of novel integrated photonic devices, including linear and nonlinear optics, quantum optics, cavity electro-optics, and piezoelectric opto-mechanics [9–12].

The preparation of LNOI substrates mainly includes two aspects: ion-slicing and wafer bonding. Ion-slicing technology plays a crucial role in the production of LNOI substrates, and revealing the mechanism of ion-slicing is beneficial to improve the quality of $LiNbO_3$ thin film. $LiNbO_3$ is highly anisotropic, while photonic and acoustic devices are manufactured on $LiNbO_3$ surfaces with different orientations [13–15]. Therefore, according to application requirements, it is necessary to prepare LNOI substrates with various orientations [16]. In general, light ions (e.g., H or He) are used for ion-slicing of $LiNbO_3$ wafers, mainly because the implantation of light ions can generate a narrow damage layer inside the target material. For the increasingly sophisticated silicon-on-insulator (SOI) technology, H and He ions implantation typically forms a defect-filled damage layer at the end of the ion range in Si [17–20]. The implantation dose of H ions is usually $4 \times 10^{16} \sim 2 \times 10^{17}$ cm^{-2}, which enables the formation of a large number of bubbles and defect clusters in the implanted damage layer [21–24]. The relevant stress leads to fractures at the corresponding depth and detachment of the Si surface layer. Similarly, ion implantation plays an important role in the formation of damaged layers during the exfoliation of $LiNbO_3$ thin films. Due to the presence of O in $LiNbO_3$, to avoid the formation of possible H–O bonds during ion implantation, He ions are used instead of H ions for ion implantation [25]. In 2019, Huang et al. investigated the ion-slicing mechanism of He ions implanted into $LiNbO_3$ chips with different orientations and found that Z-cut $LiNbO_3$ exhibited regular surface blisters and plate-like exfoliation, while Y-cut $LiNbO_3$ showed a unique "rolled-up" spalling [26]. In 2021, Jia et al. systematically calculated the distribution range of He ions in $LiNbO_3$ crystals under different implantation energies based on Monte Carlo theory. By optimizing parameters such as ion implantation uniformity, implantation time, and

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temperature, 4-inch $LiNbO_3$ thin films were prepared [27]. In 2023, Dolabella et al. revealed the mechanism of crystal recovery and gradual reduction of defect cluster size due to lattice strain and stress induced by ion-slicing in $LiNbO_3$ crystals through the study of coherent scattering and diffuse scattering by high-resolution X-ray diffraction [28]. The above research content has a certain reference value for revealing the physical mechanism of the $LiNbO_3$ wafer exfoliation process. However, the physical mechanism of surface exfoliation of $LiNbO_3$ has not been fully elucidated. Furthermore, researchers have mainly focused on improving the surface exfoliation efficiency and wafer bonding of $LiNbO_3$, but there are relatively few feasibility studies on the combination of the two aspects.

In this work, on the one hand, high exfoliation efficiency of $LiNbO_3$ thin films is achieved based on ion-slicing technology, and on the other hand, the transfer of $LiNbO_3$ films to Si-based substrates is realized based on plasma-activated bonding technology. The greatly enhanced bonding strength ensures that the interface thermal stress and diffusion of atoms during the annealing process will not cause $LiNbO_3$ films to crack. Therefore, a defect-free $LiNbO_3$ thin film on the insulator is obtained at 250 °C. Based on the characterization and analysis of the bonding interface and surface, an equilibrium system is established between ion implantation dose, interface thermal stress, and film exfoliation temperature. In addition, a physical model for the detachment of large-sized $LiNbO_3$ films by ion beam implantation is established for the first time, and the threshold conditions for the detachment of $LiNbO_3$ films are also obtained in the paper.

2. Experimental methods

Z-cut double-sided polished $LiNbO_3$ substrates and double-sided polished Si substrates were utilized in experiments for the preparation of LNOI substrates. Both $LiNbO_3$ and Si have the size of 10×10 mm^2 and the thickness of 500 μm. As shown in Fig. 1, an amorphous SiO_2 (a-SiO_2) layer was first grown on the surface of $LiNbO_3$ and Si by plasma-enhanced chemical vapor deposition (PECVD). The thickness of the a-SiO_2 layer on the surface of $LiNbO_3$ was 200 nm, and the thickness of the a-SiO_2 layer on the surface of Si was 5 μm. After deposition, the $LiNbO_3$ substrates were placed in an ion implanter, and ion implantation was performed on the $LiNbO_3$ substrate. The implantation energy of He ions was 160 keV, and the implantation doses were 2×10^{16} ions/cm^2, 4×10^{16} ions/cm^2, and 6×10^{16} ions/cm^2, respectively. After ion implantation, $LiNbO_3$ substrates were placed in an annealing furnace. The annealing temperatures were set to 200 °C, 250 °C, and 300 °C for 1 h, and samples with different annealing temperatures were obtained. The three samples were placed under an optical microscope for observation to determine the optimal ion implantation conditions and annealing temperature. Then, the a-SiO_2 layers on the $LiNbO_3$ and Si surfaces were chemically mechanically polished to reduce surface roughness to below 0.5 nm. After polishing, $LiNbO_3$ and Si substrates were placed in the plasma-activated chamber, respectively. When the vacuum inside the chamber was dropped to 40 Pa, plasma activation began. The Ar gas flow rate was 40 sccm. The implantation energy of the Ar atoms beam was 2 keV, the implantation dose was 1×10^{15} ions/cm^2, and the activation time was 90 s. After activation, the activated a-SiO_2 layers on the surfaces of $LiNbO_3$ and Si were exposed to air, and then immediately came into contact to achieve pre-bonding. With the help of an external force, the unbonded region between $LiNbO_3$ and Si was eliminated. To saturate the pre-bonding strength, $LiNbO_3$/a-SiO_2/Si bonding pairs were stored at room temperature for 24 h. Then, $LiNbO_3$/a-SiO_2/Si bonding pairs were placed in an annealing furnace for annealing to achieve the exfoliation of the upper layer $LiNbO_3$ films.

According to the simulation results of SRIM (The Stopping and Range of Ions in Matter), the concentration and implantation damage distribution of different doses of He ions with an implantation energy of 160 keV in a-SiO_2/$LiNbO_3$ substrates were shown in Fig. 2 [29]. The maximum concentration depth of He ions was 756 nm, and the maximum damage depth caused by implantation was 696 nm. The OCA-50 Data Physics instrument is used to characterize the surface hydrophilicity of a-SiO_2 layer. The surface roughness and morphology of the a-SiO_2 layer were characterized by atomic force microscopy (AFM,

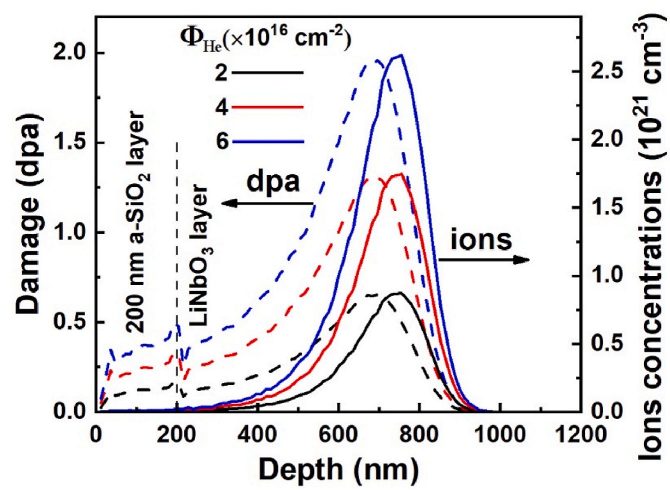

Fig. 2. Concentration and implantation damage distribution of He ions with an implantation energy of 160 keV in a-SiO_2/$LiNbO_3$ substrates.

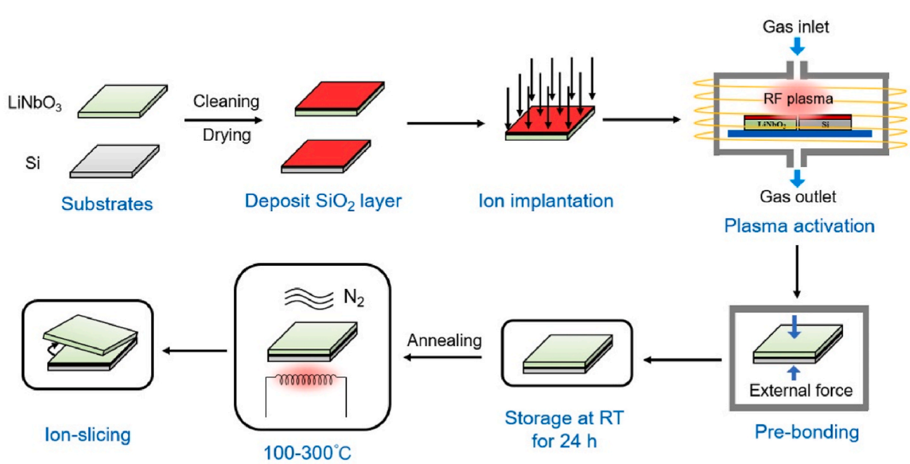

Fig. 1. Process flowchart for the preparation of $LiNbO_3$ thin films on insulators.

Dimension Icon, Bruker). The bonding strength was measured by using a tensile machine (Instron 5569, Instron). The tensile rate was fixed at 1 mm/min. The samples were thinned using a focused ion beam system (FIB, Helios 450s, FEI) to meet the needs of morphology observation. Transmission electron microscopy (TEM, Titan Themis 200, FEI) was utilized to observe the bonding interface of the samples. The distribution of interface elements was analyzed using energy-dispersive X-ray spectroscopy (EDS). The cross-sectional morphology of LNOI substrates was detected by a scanning electron microscope (Gemini SEM 300, ZEISS). Optical microscopy (OM, Olympus, Japan) was utilized to observe the surface damage of annealed $LiNbO_3$ substrates.

3. Results and discussion

3.1. Surface morphology characterization

Fig. 3(a)-3(c) show the surface morphology of the annealed $LiNbO_3$ substrates with an implantation dose of 2 × 10^{16} ions/cm^2, and the annealing time is 1 h. Fig. 3(a) shows the optical microscope image of $LiNbO_3$ surface annealed at 200 °C. It can be seen that the surface is intact and there is no blistering or exfoliation of the surface. When the annealing temperature is further increased to 250 °C, as shown in Fig. 3(b), many blisters appear on the surface. When the annealing temperature rises to 300 °C, bulk peeling fragments appear on the surface. At this temperature, bubbles intensify and coalesce together, causing the ion implantation-damaged layer to detach from the surface of the chip. Fig. 3(g)-3(i) show AFM images of $LiNbO_3$ surface with an implantation dose of 2 × 10^{16} ions/cm^2 at annealing temperatures of 200 °C, 250 °C, and 300 °C, respectively. It can be seen that as the annealing temperature increases, the number and size of blisters become larger, and the surface begins to peel off. To study in detail the effects of different implantation doses on the surface exfoliation of $LiNbO_3$ substrates, three $LiNbO_3$ chips with different implantation doses of 2 × 10^{16} ions/cm^2, 4 × 10^{16} ions/cm^2, and 6 × 10^{16} ions/cm^2 are compared. As shown in Fig. 3(d)-3(f), the surface exfoliation of $LiNbO_3$ chips with different implantation doses is observed at an annealing temperature of 300 °C for 1 h. A large number of blisters and a small amount of exfoliation appear on the surface of $LiNbO_3$ chips with an implantation dose of 2 × 10^{16} ions/cm^2, and the exfoliation size of the chip is relatively small. The surface of the $LiNbO_3$ chip with an implantation dose of 4 × 10^{16} ions/cm^2 exhibits significant plate-like exfoliation. On the surface of $LiNbO_3$ chips with an implantation dose of 6 × 10^{16} ions/cm^2, most of the $LiNbO_3$ thin films have been peeled off. It can be inferred that as the implantation dose increases, $LiNbO_3$ crystals with higher implantation

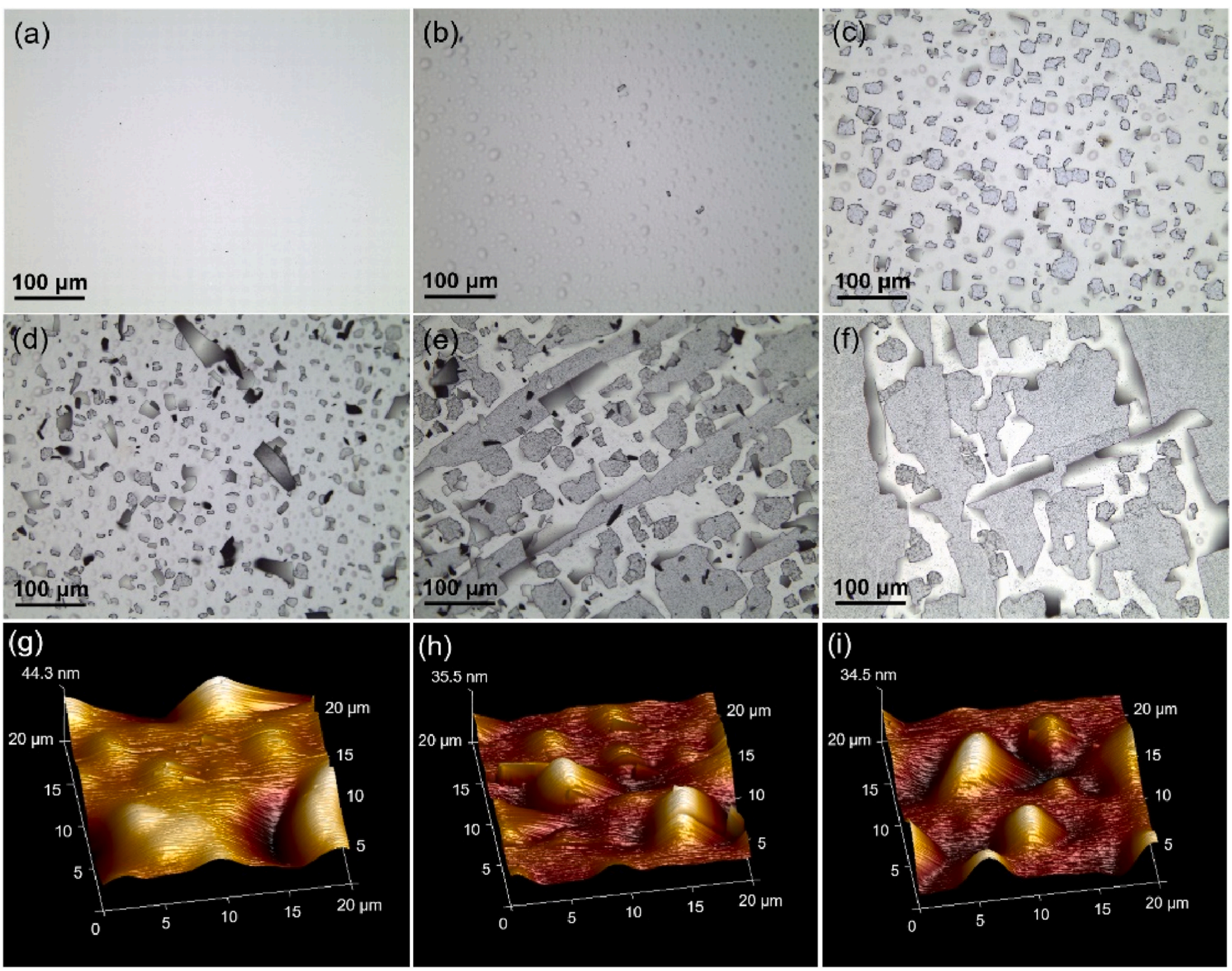

Fig. 3. (a)–(c) Surface morphology of $LiNbO_3$ chips with an implantation dose of 2 × 10^{16} ions/cm^2 annealed at 200 °C, 250 °C, and 300 °C, respectively; (d)–(f) Surface morphology of $LiNbO_3$ chips with an implantation dose of 2 × 10^{16} ions/cm^2, 4 × 10^{16} ions/cm^2, and 6 × 10^{16} ions/cm^2 after annealing at 300 °C; (g)–(i) AFM images of $LiNbO_3$ surface implanted with a dose of 2 × 10^{16} ions/cm^2 at annealing temperatures of 200 °C, 250 °C, and 300 °C, respectively.

doses are more prone to surface exfoliation at the same temperature. In addition, an increase in implantation dose can also reduce the annealing temperature for exfoliation of the $LiNbO_3$ surface. However, excessive implantation dose will introduce more lattice damage inside the $LiNbO_3$ material, reducing the quality of the $LiNbO_3$ film. Therefore, in the subsequent exfoliation experiment, we choose an implantation dose of 2 $\times$ 10^{16} ions/cm^2 for the He ions.

3.2. Characterization of cross-sectional damage caused by ion implantation

Fig. 4(a)-4(c) show the cross-sectional morphology of the sample with an implantation dose of 2 $\times$ 10^{16} ions/cm^2 after 250 °C annealing for 1 h. Fig. 4(d)-4(f) show the cross-sectional morphology of the sample with an implantation dose of 4 $\times$ 10^{16} ions/cm^2 after 250 °C annealing for 1 h. It is well known that He ion implantation can lead to a large number of nano-defects at the implantation depth. These defects migrate, accumulate, and grow into microcracks during the annealing process, forming a damaged layer with a thickness of approximately 170 nm, as shown in Fig. 4(a). The effect of the implanted He ions is to generate pressure inside the microcracks and increase the stress around the microcracks. When the stress intensity factor at the crack tip exceeds the fracture toughness of $LiNbO_3$ crystal, the crack propagates laterally, forming Type-I defects parallel to the sample surface, as shown in Fig. 4(b). In addition, $LiNbO_3$ is a highly brittle crystal, and stress in the material can cause plastic deformation and crystal cracks [6]. As shown in Fig. 4(b), Type-II defects will form a certain angle to the sample surface, which is a crystal crack caused by implantation-induced stress. During the annealing process, Type-I and Type-II defects propagate, aggregate, overlap, and ultimately form continuous cracks between the peeled $LiNbO_3$ film and the substrate, as shown in Fig. 4(f). The evolution process of defects is shown in Fig. 4(c)-4(e). Implantation and diffusion of He ions will form plate-like defects and defect clusters inside the $LiNbO_3$ crystal. As the internal pressure of the defect increases, the size of the defect further increases and eventually coalesces, forming micron-sized Type-I cracks parallel to the wafer surface. High-dose He ions implantation causes high stress and generates stress-related Type-II cracks. Type-II cracks serve as intermediates connecting Type-I cracks, accelerating the formation of cracks. In addition, cracks are prone to propagate laterally with the assistance of Type-II cracks, and the ratio of delamination height to crack length is small.

3.3. Surface roughness and wettability characterization

In the plasma-activated bonding process, surface roughness and wettability are two very important parameters. As shown in Fig. 5, before Ar plasma activation, the surface roughness of the a-SiO_2 layer on the surface of $LiNbO_3$ and Si is 0.280 nm, and the contact angle of surface droplets is 83.8°. This indicates that the surface of the a-SiO_2 layer exhibits high hydrophobicity before activation, which is consistent with previous research results [30]. After Ar plasma activation for 90 s, the surface roughness of the a-SiO_2 layer becomes 0.335 nm, and the contact angle of surface droplets is 17.4°. After activation, the surface of the a-SiO_2 layer remains smooth and flat, but the surface contact angle decreases, which means that the surface hydrophilicity is enhanced. Based on the above experimental results, Ar plasma activation can maintain the smoothness of the a-SiO_2 layer on the surface of $LiNbO_3$ and Si, enhance the hydrophilicity of the $LiNbO_3$ surface, and improve the

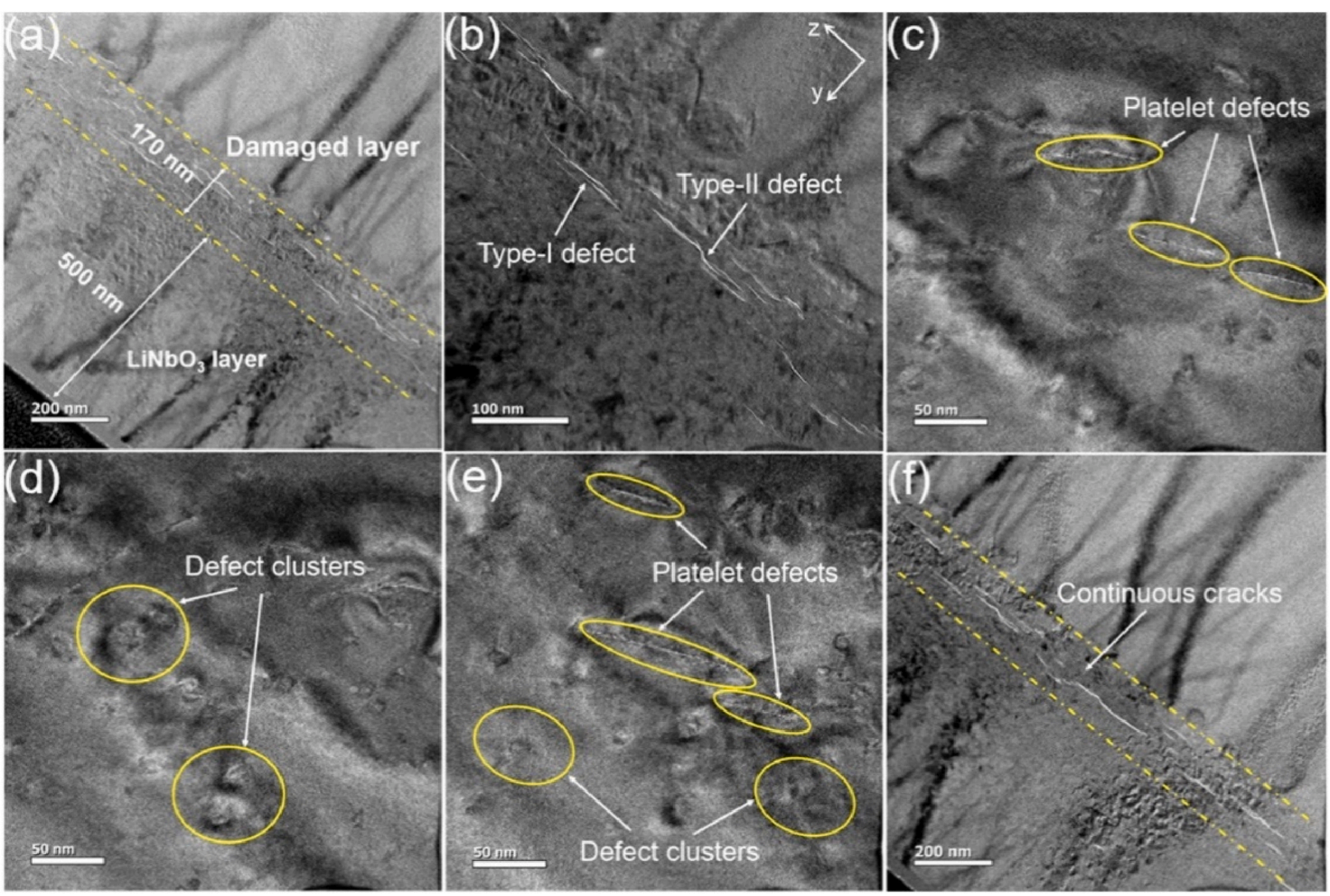

Fig. 4. The cross-sectional morphology of the sample annealed at 250 °C for 1 h. (a)–(c) The sample was implanted with a dose of 2 $\times$ 10^{16} ions/cm^2; (d)–(f) The sample was implanted with a dose of 4 $\times$ 10^{16} ions/cm^2.

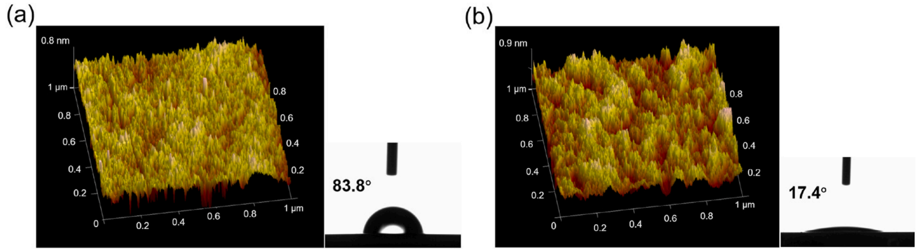

Fig. 5. Surface roughness and hydrophilicity of the a-SiO_2 layers on $LiNbO_3$ and Si surfaces before and after Ar plasma activation.

bonding strength between $LiNbO_3$ and Si substrates. Due to the size of the surface wettability, it is usually related to the density of the hydrophilic functional groups on the wafer surface. Therefore, the greater the density of the hydrophilic functional group, the more hydrophilic the surface and the smaller the wetting angle. The existing research results indicate that after Ar plasma activation, the number of surface hydroxyl functional groups will be significantly increased [31,32].

3.4. *Characterization of bonding interface and establishment of thermal stress model*

Firstly, the FIB system is used to thin $LiNbO_3$/a-SiO_2/Si bonded samples to meet the observation needs of TEM. Fig. 6 shows the interface morphology of $LiNbO_3$ and a-SiO_2 layers. From Fig. 6(a), it can be seen that the interface of the a-SiO_2 layer grown on the surface of $LiNbO_3$ through PECVD is dense and there is no defect formation. However, there are several nanometer-thickness lattice disorders on the side of $LiNbO_3$, mainly due to the impact of He ions on the surface of $LiNbO_3$ after passing through the a-SiO_2 layer. As shown in Fig. 2, at the interface between the a-SiO_2 layer and $LiNbO_3$, the damage introduced by He ions implantation is relatively greater than that on both sides, so Fig. 6 (a) shows the occurrence of lattice dislocations on the surface of $LiNbO_3$. Fig. 6(b)-6(e) show the distribution of various elements at the interface, from which it can be seen that the Si and Nb elements have mutual diffusion phenomena at the interface. From the element intensity distribution curve in Fig. 6(f), it can be seen that there is still a certain amount of C element at the interface, which is mainly caused by organic pollution.

To analyze the thermal stress generated at the interface of $LiNbO_3$/a-SiO_2/Si bonded samples during annealing, a $LiNbO_3$/a-SiO_2/Si bonding model is established based on the finite element method [33]. The thicknesses of $LiNbO_3$, a-SiO_2, and Si are 500 μm, 5 μm and 500 μm respectively. The lattice constants of $LiNbO_3$, a-SiO_2, and Si are 0.5147 nm, 0.5317 nm, and 0.565 nm, respectively. The thermal expansion coefficients of $LiNbO_3$, a-SiO_2, and Si are 14.4×10^{-6}/K, 0.5×10^{-6}/K, and 2.5×10^{-6}/K, respectively. Poisson's ratios of $LiNbO_3$, a-SiO_2, and Si are 0.25, 0.17, and 0.29, respectively. A uniform cylindrical heat source is set in the $LiNbO_3$ layer, and the heat exchange with the external area can only pass through the $LiNbO_3$ layer and the Si substrate. Adiabatic boundary conditions (no heat flow) are applied to the

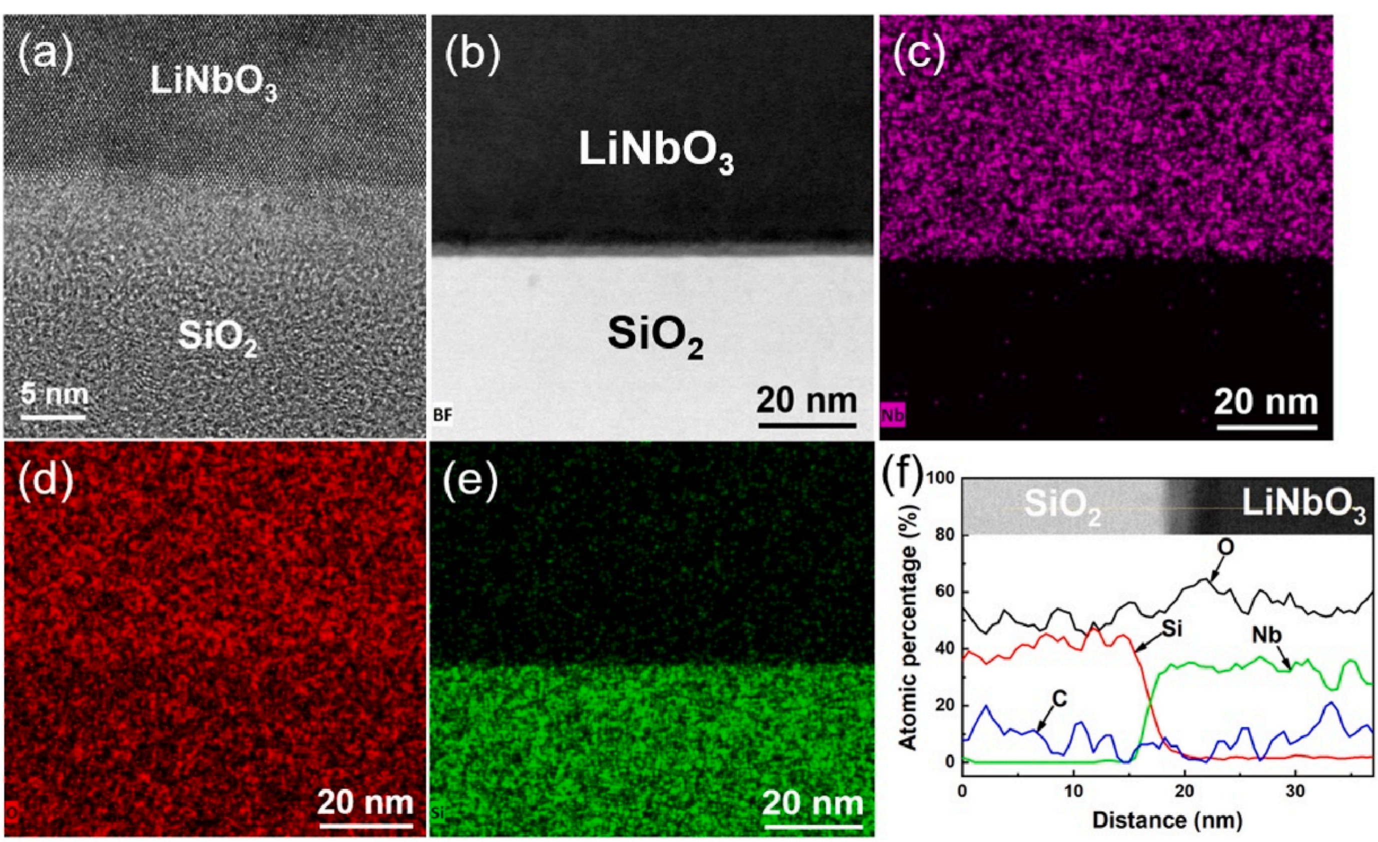

Fig. 6. Interface morphology of $LiNbO_3$ and a-SiO_2 layers. (a) and (b) The interfaces of $LiNbO_3$/a-SiO_2 with different magnifications; (c), (d), and (e) Element distribution maps of Nb, O, and Si; (f) EDS line scan curves of elements at the interface.

sidewalls and top surfaces of the model structure, and the boundary temperature conditions on the bottom surface of the Si substrate are set to maintain room temperature. Fig. 7(a) shows the z-displacement cloud map of $LiNbO_3$/a-SiO_2/Si bonded samples. The deformation and interfacial thermal stress of the bonded samples were simulated at annealing temperatures of 100 °C, 150 °C, 200 °C, 250 °C, and 300 °C, respectively. The simulation results are shown in Fig. 7(b) and (c). The results extract data along the diagonal direction of a square at the bonding interface. The four edges at the bottom of the Si substrate act as supporting edges. As the annealing temperature increases from 100 °C to 300 °C, the overall maximum deformation of the bonded chip increases gradually from 15 μm to 55 μm. The chip presents a warped state as a whole. Fig. 7(c) is a schematic diagram of the equivalent stress distribution at the interface. From the figure, it can be seen that the stress distribution in the center of the chip is relatively uniform. But at a distance of 1 mm from the edge, the stress suddenly increases and then sharply decreases. This result indicates that the thermal stress near the edge is relatively high, which can lead to the phenomenon of lack of bonding or low bonding strength at the edge. The interface deformation is convex in the middle, indicating that the middle deformation is the largest.

The bonding strength of $LiNbO_3$/a-SiO_2/Si bonding chips with different annealing temperatures is tested by a tensile measuring instrument. Bonding squares with a size of 10 × 10 mm^2 are fixed on the fixture using an epoxy adhesive, and the metal clamp is slowly moved at a speed of 1 mm/min to measure the bonding strength of different samples. Fig. 8 shows the bonding strength of $LiNbO_3$/a-SiO_2/Si samples at different annealing temperatures, with a maximum bonding strength of 4.7 MPa at 150 °C. As the annealing temperature further increases, the bonding strength begins to decrease, and at 250 °C, the bonding strength is 2.8 MPa. In the ion implantation experiment, we have determined that the implantation dose of He ions is 2×10^{16} ions/cm^2 to reduce the lattice damage inside $LiNbO_3$ material. This dose is sufficient to cause blisters and exfoliation on the surface of $LiNbO_3$ at 250 °C. Although an annealing temperature of 250 °C can cause high thermal stress at the bonding interface, a low annealing temperature cannot cause the $LiNbO_3$ material to peel off. Based on this consideration, we have determined that the implantation dose of He ions is 2×10^{16} ions/cm^2 and the annealing temperature is 250 °C.

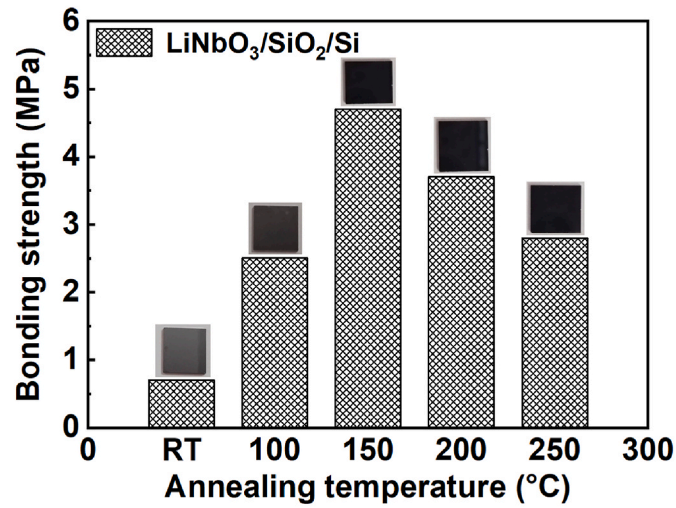

Fig. 8. Bonding strength distribution of $LiNbO_3$/a-SiO_2/Si samples at different annealing temperatures.

3.5. Preparation and application of LNOI substrates

We have determined the implantation conditions and annealing process parameters for He ions. Then, we will proceed with the exfoliation experiment of $LiNbO_3$ thin films. The $LiNbO_3$/a-SiO_2/Si bonded chips are placed in an annealing furnace and the annealing temperature is slowly increased at a rate of 1 °C/min. When the annealing temperature is raised to 250 °C and is maintained for 12 h, the upper layer of $LiNbO_3$ film peels off from the substrate and remains on the a-SiO_2 layer. After annealing, the bonded samples are removed after the temperature in the furnace has cooled naturally to room temperature. The top layer of $LiNbO_3$ film is then polished. Fig. 9(a) is a cross-sectional SEM image of the $LiNbO_3$ film on an insulator after exfoliation. The thickness of the upper $LiNbO_3$ film is 401.9 nm, the thickness of the middle a-SiO_2 layer is 4.801 μm, and the thickness of the bottom Si substrate is 500 μm. Since the a-SiO_2 layer needs to be polished before bonding, the thickness of the a-SiO_2 layer has been reduced. Fig. 9(b)-9(d) are the schematic diagram

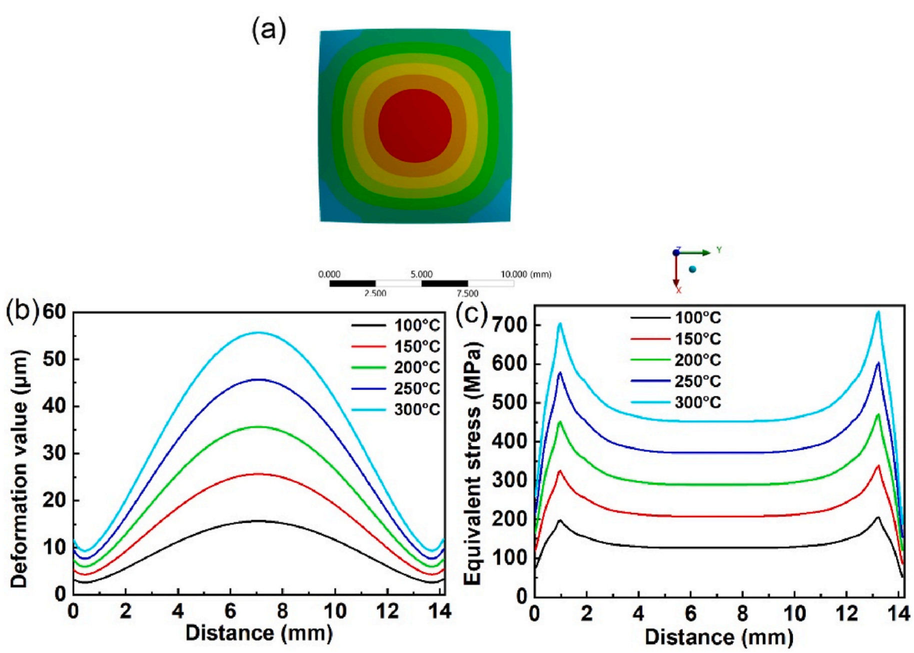

Fig. 7. The $LiNbO_3$/a-SiO_2/Si interface stress model was established based on the finite element method. (a) Z-axis displacement cloud map of the $LiNbO_3$/a-SiO_2/Si bonded sample, (b) and (c) show the deformation and interface thermal stress distribution of the bonded sample after annealing.

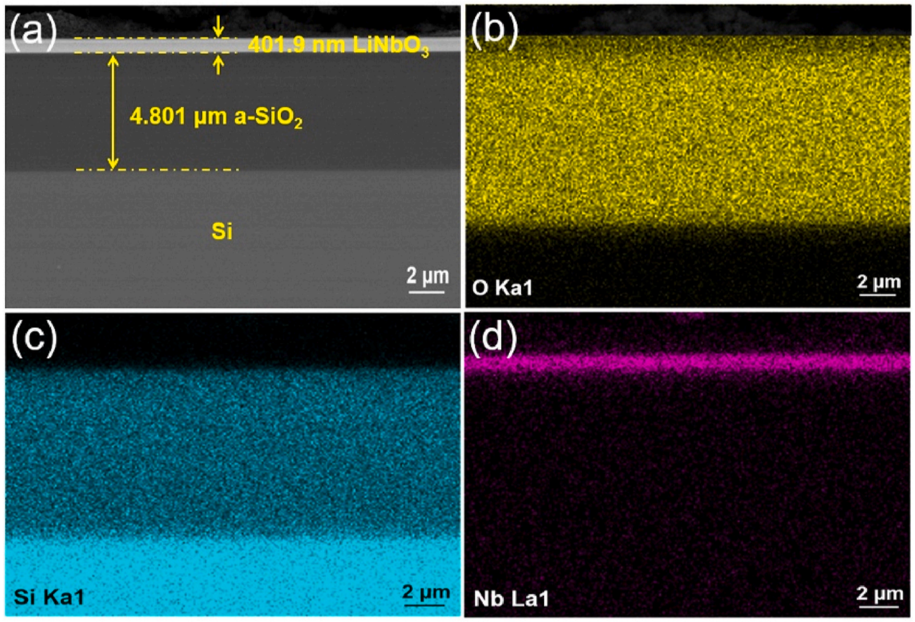

Fig. 9. (a) Cross-sectional morphology of $LiNbO_3$ thin film on insulator, (b), (c), and (d) show the distribution of O, Si, and Nb elements, respectively.

of the energy spectrum analysis of cross-sectional elements. It can also be observed from the figure that the Nb and Si elements have undergone a mutual diffusion phenomenon. This is mainly because the implanted He ions transfer part of their energy to Nb and Si atoms, causing these two elements to migrate at the interface.

To verify whether the prepared $LiNbO_3$ thin film on the insulator has practical value, optical waveguide etching is performed on the surface of LNOI [34]. Firstly, a layer of metallic Cr is sputtered on the surface of LNOI using magnetron sputtering as a mask to block the patterned area from proton exchange and wet etching. Then, the optical waveguide pattern is exposed using electron beam lithography. A mixed solution of cerium ammonium nitrate ($(NH_4)_2Ce(NO_3)_6$) and HNO_3 is used to wet etch part of the Cr mask, and the residual photoresist is removed with acetone in a water bath to transfer the mask pattern onto the metal Cr film. This is followed by a proton exchange process, in which the mixed powder with benzoic acid/lithium benzoate is removed from the furnace tube. The temperature in the furnace tube is increased to 240 °C and held for 10 min. After proton exchange is completed, the samples are washed with ethanol and deionized water after natural cooling. Subsequently, the samples are immersed in a mixed solution of hydrofluoric acid (HF) and HNO_3 at room temperature for wet etching. After the wet etching is completed, a mixed solution of $(NH_4)_2Ce(NO_3)_6$ and HNO_3, as well as ethanol solution at 70 °C, is used to remove the residual metal mask layer on the sample. Finally, the samples are washed again. Fig. 10 (a) shows the surface morphology of the sample after wet etching. From the figure, it can be observed that exfoliation has occurred in the surrounding area of the sample, mainly due to the drilling and etching problem of the a-SiO_2 layer in the middle and the excessive stress near the edge of the $LiNbO_3$ film. However, most of the middle area of the sample remains intact, as shown in the yellow line in the figure. Fig. 10 (b) shows the Y-shaped branching optical waveguide observed under an optical microscope. The width of the prepared ridge waveguide is 2 μm and the angle of the Y-shaped waveguide is 15°. The above experimental results can prove that the LNOI substrate prepared in this work has application value and can fully meet practical application requirements.

4. Conclusions

Preparation of $LiNbO_3$ thin film on insulator is realized by Ar plasma-activated bonding and ion-slicing techniques. The Y-shaped branching optical waveguide prepared by wet etching and proton exchange processes can demonstrate the practical application value of the LNOI substrates in this experiment. In addition, the mechanism of annealing temperature and ion implantation dose on the exfoliation of the $LiNbO_3$ thin film is analyzed. The implantation dose of He ions is determined to be 2×10^{16} ions/cm^2, and the annealing temperature is 250 °C. The evolution mechanism of internal defects and material exfoliation process is studied through TEM. A $LiNbO_3$/a-SiO_2/Si bonding model is established based on the finite element method. The thermal stress and deformation distribution of the interface at different temperatures are

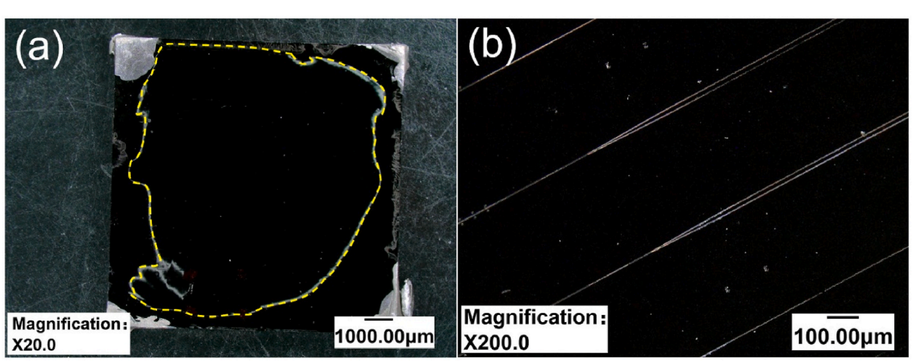

Fig. 10. (a) Surface morphology of LNOI sample after wet etching, (b) Y-shaped branching optical waveguide on the surface of LNOI sample.

obtained. The thermal stress near the edge of the bonded chip is relatively high. Based on the observation of the morphology of the $LiNbO_3$/a-SiO_2/Si bonding interface, it is found that Nb and Si atoms at the interface undergo interdiffusion due to the implantation of He ions. The interface of the prepared LNOI substrate is defect-free and firmly bonded. The excellent material properties can prove that the LNOI substrates prepared in this study can fully meet the demanding requirements of high-performance thin film $LiNbO_3$ devices.

CRediT authorship contribution statement

Rui Huang: Writing – original draft, Software, Resources, Project administration, Formal analysis, Conceptualization. **Xin Zhang:** Software, Resources. **Mingzhi Tang:** Methodology, Data curation. **Rui Li:** Methodology. **Hao Xu:** Software, Resources. **Yecai Guo:** Supervision. **Zhiyong Wang:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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